Bharath Mandha

Applied AI / ML Engineer | 4+ years in applied ML, 1+ year in production retrieval and LLM systems
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EXPERIENCE

Goldman Sachs, *AI Engineer*, *Dallas, TX* Jun 2025 – Present

- Owned the retrieval layer, LangGraph orchestration, FastAPI integration, evaluation harness and ingestion of a RAG platform over 3M+ internal financial documents (10M+ chunks), as one of 3 ML engineers on a 7-person team; about 73 credit and risk analysts use it weekly
- Analysts spent 2.9 hours preparing each of 112 platform-assisted credit memos, down from 4.8 hours across 106 historical ones, about 40% less. That's an observational comparison, not a randomized one
- Scored every retrieval change on one 520-query benchmark kept apart from generation and tuned on a separate dev set. Recall@10 rose from 84.6% to 91.7% over the dense FAISS baseline in four stages (filtering +1.5 points, chunking and indexing +1.9, hybrid retrieval +2.2, a BGE cross-encoder over the top 30 +1.5), MRR from 0.73 to 0.84, and p95 retrieval latency fell from 1.42 s to 860 ms; RAGAS groundedness stayed above 90%
- Built the LangGraph agents that completed 492 of 784 targeted workflow steps (62.8%) across 112 platform-assisted memos, from retrieval through citation assembly and draft sections; a step counts only if its output passed validation and the analyst didn't redo it. Final credit judgments stayed with analysts
- Wrote the design for the evaluation framework and incremental indexing (versioned benchmarks, regression thresholds, index-consistency checks), presented it in design reviews and helped two engineers adopt it; it became the shared way to compare retrieval, reranking and orchestration changes
- Tested quantization and LoRA/QLoRA on Hugging Face models for quality, latency, GPU memory and cost, and recommended keeping the existing model because the gains did not justify the move

Cigna Healthcare, *Machine Learning Engineer*, *Remote, India* Mar 2022 – Aug 2023

- Raised 30-day readmission ROC-AUC from 0.846 (regularized logistic regression) to 0.912 with XGBoost and LightGBM over 1.5M+ encounters, on a held-out cohort of about 218K encounters (about 72K patients) with time-aware, patient-level splits
- In an offline test on the held-out cohort with outreach capped at 1,000 patients, the model's top 1,000 held 245 actual readmissions, where baseline prioritization caught 213 (+15%, precision 24.5% vs 21.3%)
- Owned feature work (150+ candidates, about 100 kept), MLflow tracking, PSI drift checks and retraining criteria, and deployed on SageMaker and EKS with platform engineers. SHAP explanations supported clinical review

Deloitte, *Associate Data Scientist*, *Hyderabad, India* Jun 2020 – Feb 2022

- Cut weighted MAPE on SKU- and store-level demand forecasts for a retail client from 24.8% to 21.3% (14% relative), using gradient-boosted models and rolling-origin backtests against the established baseline
- Built the PySpark, SQL and BigQuery pipelines behind it, over 50M+ transactions

PROJECTS

wiki-retrieval, *retrieval over 2.68M passages* | github.com/bharathkumardev1/wiki-retrieval | demo: youtu.be/5vtW4iaUe3A

- Fine-tuned MiniLM with LoRA for dense retrieval over the full 2.68M-passage BEIR NQ corpus: nDCG@10 0.402, up 0.014 on the untuned encoder (paired p 0.0001) and 0.112 above my own BM25 (0.290). On a 104K-passage dev slice, an INT8 ONNX cross-encoder over the top 20 added 0.0941 nDCG@10 at 794 ms p50 on 2 vCPU
- Built a script-only LangGraph cite-or-refuse agent: a cross-encoder gate cut answers from 251 to 58 of 300 and raised cited-passage relevance from 69.7% to 94.8%, mostly by answering fewer questions (in-sample, one annotator)

tlc-demand, *demand forecasting and serving* | github.com/bharathkumardev1/tlc-demand | demo: youtu.be/Hd28gbz9jy0

- LightGBM Poisson forecast of 15-minute pickups in 262 NYC zones, 45 minutes past the last observed data: WAPE 0.2415, 22% below the best simple baseline (it loses in the two quietest zone quartiles)
- Java Spring Boot gateway with a circuit breaker and static fallback, run against the live ONNX service: with the model killed, 40 of 40 requests came from the fallback; PSI drift monitor raised 36 false alarms in 1,776 clean-data checks

finance-rag-eval, *retrieval over SEC filings* | github.com/bharathkumardev1/finance-rag-eval

- Page-level retrieval evaluation on FinanceBench (150 questions, 53,399 pages of SEC filings) with a held-out split and bootstrap intervals: a company and period filter (+0.29) and statement-page routing (+0.28) are most of the gain from 0.14 to 0.71 Hit@10
- A MiniLM hybrid led on the 50-question dev split but scored 0.73 against BM25's 0.75 on test at about five times the latency, so BM25 ships; one core serves about 126 queries a second

SKILLS

Languages: Python, SQL, PySpark; Java and Go for services and tooling
ML and LLMs: PyTorch, scikit-learn, LightGBM, Hugging Face, LoRA/QLoRA, LangGraph, FAISS, ONNX, RAGAS; GPU training (T4)
MLOps and cloud: MLflow, Airflow, FastAPI, Docker, Kubernetes, Terraform, CI/CD, AWS (SageMaker, S3, EKS), GCP (BigQuery)

EDUCATION

University of North Texas, MS Computer Science Aug 2023 – May 2025